# Psychology 3310: Specialist Topics
# Emotion Science



Emotion. What is it? This specialist topic will look at what emotions are, and how they might differ from moods and feelings. We will discuss how people display emotion, particularly via the face but also by the body and voice. In addition to examining emotion in everyday life, we will also examine a variety of disorders in which emotion processing is disrupted.

The science of emotion is an exciting and broad area of psychological inquiry. My aim is to expose you to some of the important and interesting questions that have been examined, using diverse methods. The seminar series will draw from a wide array of research from psychology and neuroscience.

Emotion lies at the heart of many professional applications of psychology, including clinical psychology, health psychology, and organisational psychology. Emotions also play an important role influencing decision-making in an array of ethical, legal and political domains.

The research project will allow students to investigate a novel question in emotion science, introducing some of the techniques used to measure emotion and providing experience in conducting research in this field.

## Instructors

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## Seminars (Thursday 9-11 am, Rm 3.14)

The seminar classes revolve around group discussion and student presentations. In the first seminar I will provide a quick introduction to the issues and the topics we will cover during the semester. In subsequent weeks seminars will revolve around discussion of a central paper, plus there will be presentations of 2 or 3 additional papers by individual students.

It's possible that discussing emotions or psychological disorders may make some people feel uncomfortable. If you think that this might be the case for you please let the instructors know before class. Of course, you are always free to excuse yourself from class (but please let us know that you are okay afterwards and/or seek assistance from the UWA Counselling and Psychological Service).

In Seminar 6 (week 8), current students and post-docs will come to the seminar class to present short talks on their research projects. This session also provides a great opportunity to ask students and post-docs questions about further study (e.g., Honours).

**Key readings**

For each seminar there are **one or two key readings** (see schedule below) **that all students will be expected to read prior to class**. These readings are all listed in the Schedule at the end of this document and will be available on LMS. The readings vary considerably in terms of style and difficulty. Some are easy-to-read reviews or introductions, some are research papers and some are in-depth reviews. Usually when there are two papers they are both very short. We discuss these articles in class.

All required readings will be available on LMS. The readings required for each week will be assigned in the seminars.

**Student Seminar Presentations**

In addition, for each seminar there are at least **2 further readings,** that provide a more detailed look at some aspect of the topic. **Each week a different pair (or trio) of students will be required to read these articles (or a subset, depending on how many students are allocated to each week) and present them to the whole class**. These readings are listed at the end of this document and will be available on LMS. Each student will take the role of presenter at least once during the course. There are **8 weeks during which students will make presentations to the class**. We will sort out who will present each topic during the first seminar class.

The format for presentations may vary from topic to topic and students are welcome to be creative. Students can present as a team or individually. However, presentation of each article should be about 5 minutes long. These articles vary considerably in content and length, some being reviews, some presenting a single experiment and some presenting multiple experiments. In deciding what to include and exclude in presenting your paper you should exercise your judgement about what **you think is important** for your classmates to know about. It is not necessary to present every experiment in a multi-experiment paper, for example, nor every kind of analysis etc. However, make your decisions based on a complete reading of the article (rather than just presenting the first experiment because you ran out of time to read the rest of the article, for example!) and you are welcome to consult me (Romina) about what to include.

Also, remember that this course has a strong research focus. So you need to give the class some idea about HOW the conclusions were reached e.g. what variables did the researchers manipulate? what task did the participants complete? what was the pattern of results and how did this relate to what was hypothesized? To be able to present this kind of information you will need to read the Method & Results sections of research papers, not just the Introduction & Discussion! Seriously ;-) So your presentation should give your classmates a sense of what was done and what the data were like, not just what conclusions were drawn.

**Attendance**

Students are expected to attend all seminars and labs/tutes. The small size of the seminar class means failure to attend or prepare for class has an adverse impact on the class experience of all students. Likewise, for the labs/tutes, particularly in the first half of the semester when you will be working in small groups on a group assessment (research proposal). As a courtesy, students should

inform me (or Yong) if illness, or other unforeseen circumstances, prevent them from attending a class, ahead of time where possible.

**Participation Marks (10%)**

These will be based on contributions throughout the course, including labs, and on your presentations. Failure to regularly attend seminars or labs/tutes will obviously impact your participation mark adversely.

## Labs/Tutorials (Friday 9-11 am, Rm 2.20)

The labs are devoted to devising, conducting and writing up a research project. We will guide you in developing, conducting and writing up your research projects.

## Assessment

Assessment procedures are described in the 3310 unit outline on LMS. Make sure you familiarise yourself with the unit outline, as well as the School of Psychological Science policies on plagiarism and extensions. The information in that outline supersedes all information in this topic outline.

The assessment procedures are summarised in the following table. Only those students who attend on a regular basis, give an oral presentation, and submit their data and consent forms are eligible for participation marks. For further information, consult the unit outline

| Assessment | Size | Deadline (11.45 PM) | Marker | Type | % |
|---|---|---|---|---|---|
| **Group Work 1** (Proposal) | 1200 words | Monday 4th September (Week 6) | Romina | Group | 15% |
| **Group Work 2** (Ethics) | N/A | Monday 4th September (Week 6) | | Group | 5% |
| **Group Work 3** (Presentation) | N/A | 23-27 October (Week 12) | Tutor & students | Group | 10% |
| **Written Assignment** (Report) | 2500 words | Monday 30th October | Yong | Individual | 60% |
| **Participation** | N/A | | Romina & Yong | Individual | 10% |

## Seminar Readings

** Class Reading. It is expected that everyone in the class will have read this paper (or one of the papers for Seminar 2).

**Seminar 1. Week 2: Introduction**

** Niedenthal, P.M. & Ric, F. (2017). Chapter 1: Theories of Emotion (pp.1-24). In *Psychology of emotion: interpersonal, experiential, and cognitive approaches.* (2nd Edition). New York: Routledge.

**Seminar 2. Week 3: Different views from leaders in the field**

** Barrett, L. F. (2017). The theory of constructed emotion: an active inference account of interoception and categorization. *Social Cognitive and Affective Neuroscience, 12(1)*, 1-23. doi:10.1093/scan/nsw154

OR

** Adolphs, R. (2017). How should neuroscience study emotions? by distinguishing emotion states, concepts, and experiences. *Social Cognitive and Affective Neuroscience, 12(1)*, 24-31. doi:10.1093/scan/nsw153

*Additional papers for presenters*
- Adolphs, R. (2017). Reply to Barrett: affective neuroscience needs objective criteria for emotions. *Social Cognitive and Affective Neuroscience, 12(1)*, 32-33. doi:10.1093/scan/nsw155
- Barrett, L. F. (2017). Functionalism cannot save the classical view of emotion. *Social Cognitive and Affective Neuroscience, 12(1)*, 34-36. doi: 10.1093/scan/nsw154

**Seminar 3. Week 4. Displaying emotion: faces.**

** Palermo R, O'Connor KB, Davis JM, Irons J, McKone E (2013) New Tests to Measure Individual Differences in Matching and Labelling Facial Expressions of Emotion, and Their Association with Ability to Recognise Vocal Emotions and Facial Identity. *PLoS ONE 8(6)*: e68126. doi:10.1371/journal.pone.0068126

*Additional papers for presenters*
- Ewing, L., Karmiloff-Smith, A., Farran, E. K., & Smith, M. L. (2017). Developmental changes in the critical information used for facial expression processing. *Cognition, 166*, 56-66. doi:http://dx.doi.org/10.1016/j.cognition.2017.05.017
- Willis, M. L., Murphy, J. M., Ridley, N. J., & Vercammen, A. (2015). Anodal tDCS targeting the right orbitofrontal cortex enhances facial expression recognition. *Social Cognitive and Affective Neuroscience, 10*(12), 1677-1683. doi:10.1093/scan/nsv057
- Yan, X., Andrews, T. J., & Young, A. W. Cultural similarities and differences in perceiving and recognizing facial expressions of basic emotions. *Journal of Experimental Psychology: Human Perception and Performance, 42(3),* 423-440. http://dx.doi.org/10.1037/xhp0000114

**Seminar 4.Week 5: Displaying emotion: bodies and voices**

** Schirmer, A. & Adolphs, R. (2017). Emotion perception from face, voice and touch: comparisons and convergence. *Trends in Cognitive Sciences, 21(3),* 216-228. https://doi.org/10.1016/j.tics.2017.01.001

*Additional papers for presenters*
- Atkinson, A. P., Dittrich, W. H., Gemmell, A. J., & Young, A. W. (2004). Emotion Perception from Dynamic and Static Body Expressions in Point-Light and Full-Light Displays. *Perception, 33*(6), 717-746. doi:10.1068/p5096
- Willis, M. L., Palermo, R., & Burke, D. (2011). Judging approachability on the face of it: the influence of face and body expressions on the perception of approachability. *Emotion, 11*(3), 514-523. doi:10.1037/a0022571
- Aviezer, H., Trope, Y., & Todorov, A. (2012). Body cues, not facial expressions, discriminate between intense positive and negative emotions. *Science*, *338*(6111), 1225–1229. doi:10.1126/science.1224313.

**Seminar 5. Week 7: Universality/What are the functions of emotions?**

** Ekman, P., & Cordaro, D. (2011). What is Meant by Calling Emotions Basic. *Emotion Review, 3*(4), 364-370. doi:10.1177/1754073911410740

*Additional papers for presenters*
- Crivelli, C. A.-O., Russell, J. A.-O., Jarillo, S. A.-O., & Fernandez-Dols, J. A.-O. (2017). Recognizing spontaneous facial expressions of emotion in a small-scale society of Papua New Guinea. *Emotion, 17(2)*, 337-347.
- Rychlowska, M., Jack, R. E., Garrod, O. G. B., Schyns, P. G., Martin, J. D., & Niedenthal, P. M. (2017). Functional Smiles: Tools for Love, Sympathy, and War. *Psychol Sci*, 0956797617706082. doi:10.1177/0956797617706082
- Susskind, J. M., Lee, D. H., Cusi, A., Feiman, R., Grabski, W., & Anderson, A. K. (2008). *Expressing fear enhances sensory acquisition. Nature Neuroscience, 11(7)*, 843–850. doi:10.1038/nn.2138.

**Seminar 7. Week 9: Disorders involving Emotion 1**

** Cook, R., Brewer, R., Shah, P., & Bird, G. (2013). Alexithymia, Not Autism, Predicts Poor Recognition of Emotional Facial Expressions. *Psychol Sci, 24*(5), 723-732. doi:10.1177/0956797612463582

*Additional papers for presenters*
- Golan, O., Sinai-Gavrilov, Y., & Baron-Cohen, S. (2015). The Cambridge Mindreading Face-Voice Battery for Children (CAM-C): complex emotion recognition in children with and without autism spectrum conditions. *Mol Autism, 6*(1), 22. doi:10.1186/s13229-015-0018-z
- Trevisan, D. A., Bowering, M., & Birmingham, E. (2016). Alexithymia, but not autism spectrum disorder, may be related to the production of emotional facial expressions. *Mol Autism, 7*(1), 46. doi:10.1186/s13229-016-0108-6

**Seminar 8. Week 10: Disorders involving Emotion 2**

** Marsh, A. A., PhD., Finger, E. C., M.D., Mitchell, D. G. V., PhD., Reid, M. E., Sims, C., Kosson, D. S., PhD., . . . Blair, R. J. R., P. (2008). Reduced amygdala response to fearful expressions in children and adolescents with callous-unemotional traits and disruptive behavior disorders. *The American Journal of Psychiatry, 165*(6), 712-20.

*Additional papers for presenters*
- Attwood, A. S., Easey, K. E., Dalili, M. N., Skinner, A. L., Woods, A., Crick, L., . . . Munafò, M. R. (2017). State anxiety and emotional face recognition in healthy volunteers. *Royal Society Open Science, 4*(5).
- Penton-Voak, I. S., Munafò, M. R., & Looi, C. Y. (2017). Biased Facial-Emotion Perception in Mental Health Disorders: A Possible Target for Psychological Intervention? *Current Directions in Psychological Science, 26*(3), 294-301. doi:10.1177/0963721417704405

**Seminar 9. Week 11: Sleep and Emotion**

** Killgore, W. D. S., Balkin, T. J., Yarnell, A. M., & Capaldi, V. F. (2017). Sleep deprivation impairs recognition of specific emotions. *Neurobiology of Sleep and Circadian Rhythms, 3*, 10-16. doi:http://dx.doi.org/10.1016/j.nbscr.2017.01.001

*Additional papers for presenters*
- Bennion, K. A., Payne, J. D., & Kensinger, E. A. (2017). Residual effects of emotion are reflected in enhanced visual activity after sleep. *Cognitive, Affective, & Behavioral Neuroscience, 17*(2), 290-304. doi:10.3758/s13415-016-0479-3
- Soehner, A. M., Kaplan, K. A., Saletin, J. M., Talbot, L. S., Hairston, I. S., Gruber, J., . . . Harvey, A. G. (2017). You'll feel better in the morning: slow wave activity and overnight mood regulation in interepisode bipolar disorder. *Psychological Medicine.*

**Seminar 10. Week 12: Emotional Memory**

** Kalenzaga, S., Lamidey, V., Ergis, A. M., Clarys, D., & Piolino, P. (2016). The positivity bias in aging: Motivation or degradation? *Emotion*, *16*(5), 602–610. doi:10.1037/emo0000170.

*Additional papers for presenters*
- Kensinger, E. A. (2009). Remembering the Details: Effects of Emotion. *Emotion Review, 1(2)*, 99-113. doi:10.1177/1754073908100432
- Laney, C., & Loftus, E. F. (2008). Emotional content of true and false memories. *Memory, 16*(5), 500-516. doi:10.1080/09658210802065939